

# Suyeon Lee

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## RESEARCH INTEREST

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My research focuses on enhancing conversational and auditory experiences in challenging acoustic environments. I am also interested in exploring generative models and multimodal contrastive learning.

## EDUCATION

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| <b>Korea Advanced Institute of Science and Technology (KAIST)</b><br><i>Ph.D. in Electrical Engineering</i>                                                                                                       | Mar 2025 - Feb 2028 (Expected)<br><i>Advisor: Prof. Joon Son Chung</i> |
| <b>Korea Advanced Institute of Science and Technology (KAIST)</b><br><i>M.S. in Electrical Engineering</i> <ul style="list-style-type: none"><li>Thesis: Multi-modal Speech and Audio Separation System</li></ul> | Mar 2023 - Feb 2025<br><i>Advisor: Prof. Joon Son Chung</i>            |
| <b>Korea Advanced Institute of Science and Technology (KAIST)</b><br><i>B.S. in Electrical Engineering</i>                                                                                                        | Mar 2018 - Feb 2023                                                    |

## EXPERIENCE

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| <b>Undergraduate Research Intern</b><br><i>Multimodal AI Lab, KAIST</i> <ul style="list-style-type: none"><li>Acquired foundational knowledge in generative models, specifically exploring diffusion-based models</li><li>Implemented active speaker detection baselines using TalkNet-ASD</li></ul> | Mar 2022 - Feb 2023<br><i>Advisor: Prof. Joon Son Chung</i> |
| <b>Research Intern</b><br><i>AIRS Company, Hyundai Motor Group</i> <ul style="list-style-type: none"><li>Developed wake word detection baseline model for in-car voice assistant</li></ul>                                                                                                           | Sep 2021 - Feb 2022<br><i>Seoul, South Korea</i>            |
| <b>Undergraduate Research Intern</b><br><i>SSSC Lab, KAIST</i> <ul style="list-style-type: none"><li>Applied fundamental machine learning principles to develop a baseline speech recognition model utilizing PyTorch.</li></ul>                                                                     | Mar 2021 - Aug 2021<br><i>Advisor: Prof. Hoirin Kim</i>     |

## HONORS AND AWARDS

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| <b>NIST Speaker Recognition Evaluation (SRE)</b><br><i>1st Place in the Audio Track and 4th Place in the Audio-Visual Track</i> | 2024<br><i>San Juan, Puerto Rico</i> |
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## SKILLS

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| <b>Programming:</b> PyTorch, Python, C |
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## ACADEMIC ACTIVITIES

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| <b>Teaching Assistant</b><br><i>KAIST - EE488: Deep learning for Computer Vision</i> <ul style="list-style-type: none"><li>Supported students in coding assignments by offering one-on-one debugging and technical guidance</li></ul> | Fall 2023, Fall 2025 |
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## LANGUAGES

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| <b>Korean:</b> Native      |
| <b>English:</b> Proficient |

- [6] **Plug-and-Steer: Decoupling Separation and Selection in Audio-Visual Target Speaker Extraction**  
Doyeop Kwak\*, **Suyeon Lee\***, Joon Son Chung  
*Submitted to Interspeech 2026*
- [5] **Cinematic Audio Source Separation Using Visual Cues**  
Kang Zhang\*, **Suyeon Lee\***, Arda Senocak, Joon Son Chung  
*IEEE / CVF Computer Vision and Pattern Recognition Conference 2026*
- [4] **Model-Guided Dual-Role Alignment for High-Fidelity Open-Domain Video-to-Audio Generation**  
Kang Zhang\*, Trung X. Pham\*, **Suyeon Lee**, Axi Niu, Arda Senocak, Joon Son Chung  
*Conference on Neural Information Processing Systems 2025*
- [3] **FlowAVSE: Efficient audio-visual speech enhancement with conditional flow matching**  
Chaeyoung Jung, **Suyeon Lee**, Ji-Hoon Kim, Joon Son Chung  
*Interspeech 2024*
- [2] **Seeing through the conversation: Audio-visual speech separation based on diffusion model**  
**Suyeon Lee\***, Chaeyoung Jung\*, Youngjoon Jang, Jaehun Kim, Joon Son Chung  
*IEEE International Conference on Acoustics, Speech, and Signal Processing 2024*
- [1] **TalkNCE: Improving active speaker detection with talk-aware contrastive learning**  
Chaeyoung Jung\*, **Suyeon Lee\***, Kihyun Nam, Kyeongha Rho, You Jin Kim, Youngjoon Jang, Joon Son Chung  
*IEEE International Conference on Acoustics, Speech, and Signal Processing 2024*